

Error signals in L5 apical dendrites, as a TD-learning agent

Simulates mice learning a go/no-go texture task. Top-down “saliency” onto apical dendrites multiplies somatic gain, so neurons carrying task-relevant stimuli get amplified while 200 distractors stay quiet.

~2.6k

lines, Python + MATLAB

~1.5 GB

pickled results (LFS)

PIPELINE · DRIVEN BY MAIN.PY

1 · `python main.py simulate`
`simulations.run()`

10 seeds × 3 conditions: default (1800 trials), apical inhibition (4000, apical off for first 1800), mixed selectivity at $\theta_0 = 1...8$.

2 · `results/*.pickle` → `Smith/*.mat`
`save_trial_outcomes_matlab()`

Per-trial dicts pickled; the correct/incorrect vectors are exported for MATLAB.

3 · **MATLAB** · `Smith/getperfs.m`

Smith et al. (2004) state-space EM

Learning curve and learning / expert trial per seed → `performances.mat`. Only external step.

4 · `perf, then plot`
`load_smith_perf()` · `panels.plot_all()`

Merges curves back into the pickles, then renders 19 panels for Fig. 4 and Supp. 11–13 to panels/.

One trial, 18 time steps at 4 Hz



`x_som = g · x_bas` Apical input $w_{ap} \cdot x_{pre}[t] \rightarrow \text{sigmoid} \rightarrow \text{gain } (1...10\times)$ on basal drive.

`dV̂ = w · x_som` Linear TD critic with eligibility trace z ($\gamma = 1$).

`Δw_ap ∝ x_pre(x_som − θ)` BCM-like rule, clamped to $[0,1]$; $x_{pre}[t]$ tracks $|d\hat{V}|$ at each step.

`|r − r̂|` Outcome saliency drives apical activity; the actor's REINFORCE-style update is gated by it.

Modules

<code>main.py</code>	CLI dispatch: <code>plot · simulate · perf</code>	49
<code>15apical/helper.py</code>	Constants, result keys, Outcome / Simulation enums, transfer functions, paths	172
<code>15apical/simulations.py</code>	Agent (actor), <code>get_params</code> , <code>simulate_seed</code> — the whole model in one loop	397
<code>15apical/panels.py</code>	Figure code: loaders, alignment, stats, 19 panel functions	1505
<code>15apical/parula.py</code>	MATLAB parula colormap data	67
<code>Smith/*.m</code>	Third-party learning-curve estimator (EM, forward/backward filter)	448

Stack pinned in `requirements.txt`: Python 3.10, torch 1.13, numpy 1.23, scipy, matplotlib, statsmodels. No tests, no CI.